Careers in SynBio

Kadambari Bhide and Husayn Bootwala
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Introduction

In this session, we'll cover-

- How we got into SynBio
- Where SynBio is used and what roles exist
- Case studies
- Challenges and open opportunities

Kadambari Bhide

1st-year PhD, Chemical Engineering, MIT

Founding member of iGEM IITB, wet lab and bioprocess

Husayn Bootwala

PhD student, WashU

Founder of iGEM IITB, now working on biomanufacturing



Our Journey and Experience

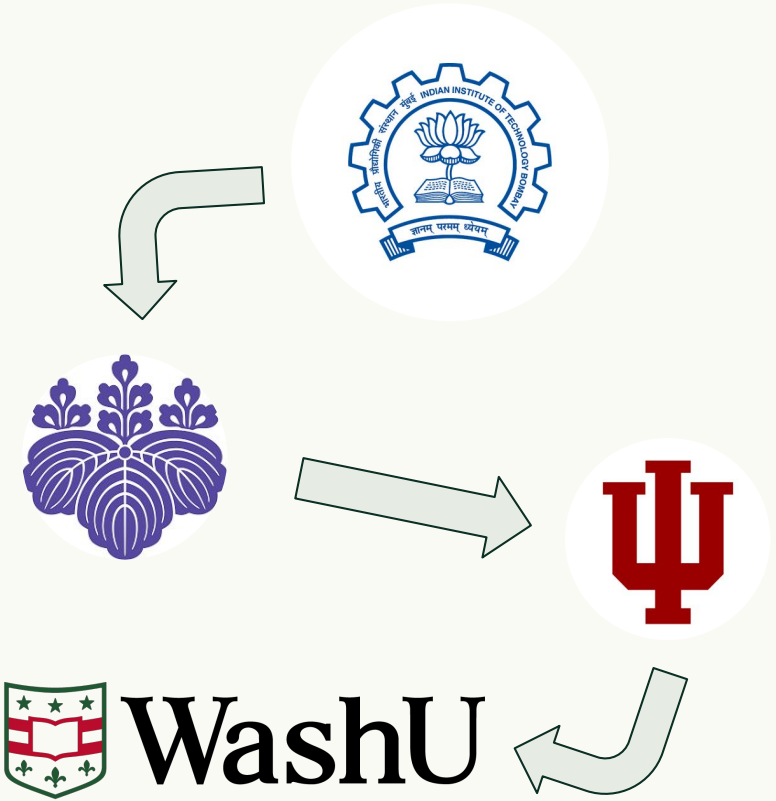
Husayn Bootwala

IITB: ChemE with a minor in BSBE; founder of iGEM IITB

Internships: Protein engineering and de novo design of biocatalysts

Exchange: University of Tsukuba; antibodies against tumor cells in mouse models

Now: 1st year PhD at WashU, working on biomanufacturing and synthetic biology



Our perspective reflects our own experience.

Our Journey and Experience

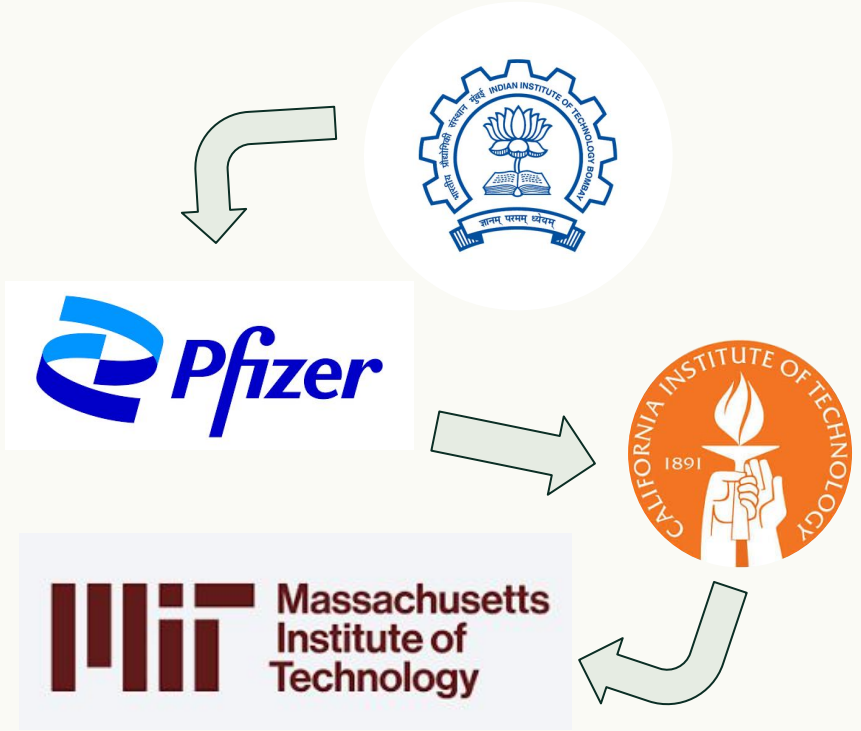
Kadambari Bhide

IITB: Chemical Engineering with a minor in BSBE

Internships: Bioprocess R&D at Pfizer, Chennai
SURF at Caltech, Manthiram lab

BTP: Bioprocess optimisation for chemostat setups, with Prof. Wangikar

Now: 1st-year PhD, Chemical Engineering, MIT

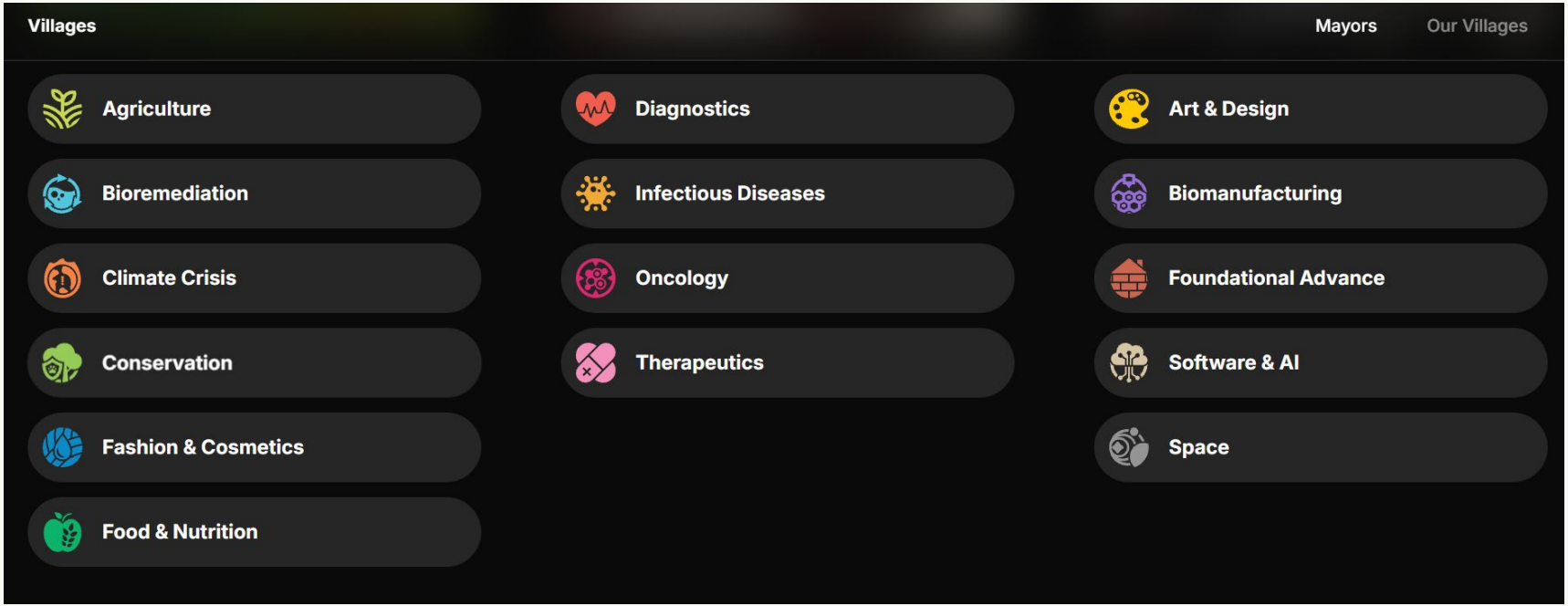


Our perspective reflects our own experience.

SynBio as a tool with diverse applications

 **Medical and healthcare**	Gene and cell therapies, microbial therapeutics, and yeast platforms that target cancers and make drugs
 **Agriculture and food**	Crops redesigned for pest and climate resilience
 **Industrial and material biosynthesis**	Microbes that make zero-carbon biofuels, bulk chemicals and biodegradable bioplastics
 **Environment and conservation**	Biosensors that detect toxins, and microbes that clean up heavy metals and oil spills

iGEM Villages



In SynBio careers, where you apply the tool matters as much as the tool

The tool

Cloning, genetic circuits, modelling and automation

+

The application

The problem you care about: a disease, a crop, a material or a climate issue

=

Your career

The role, employer and path that fit both

Know where you want to apply SynBio, and career planning gets easier

Research is not the only path: SynBio is active in academia and industry

Academia

Roles: PhD student, postdoc, professor

Work: open research questions, published as papers

Entry: MS or PhD, then a postdoc

Industry

Roles: scientist, process engineer, product developer

Work: scaling and manufacturing real products

Example: Reliance's algae-to-oil work at Jamnagar uses algae, sunlight and refinery CO2 to make bio-crude

Source: SynBioBeta; Algenol and Reliance announcement, January 2015

Computational and experimental roles do different work on the same problems

Computational roles (dry lab)

- Genomic and sequencing data with bioinformatics
- Design and optimize genetic circuits
- Modelling
- Predict system behavior with simulation, statistics and machine learning

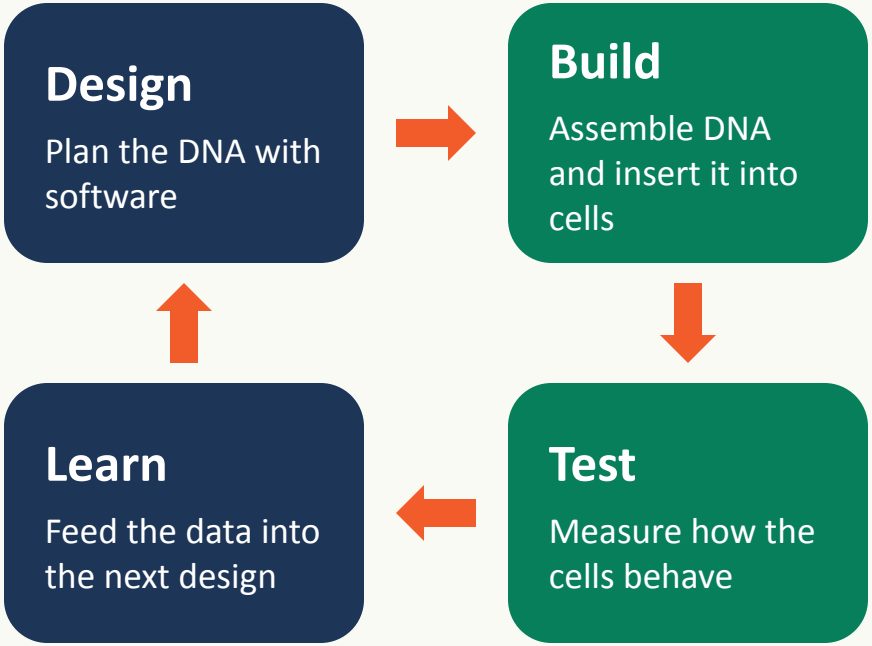
Experimental roles (wet lab)

- Engineer and genetically modify organisms and cells
- Build and test synthetic genetic circuits
- Characterize systems with flow cytometry, fluorescence assays, microscopy and molecular biology assays

They overlap: every project runs through the design-build-test-learn cycle

The two meet in the design-build-test-learn cycle

- **Design** and **Learn** are mostly dry-lab work
- **Build** and **Test** are mostly wet-lab work
- Most projects move between both in **every cycle**



Case Study: Anthropic has a Life Sciences team

Research Associate, Biology

Entry-level bench role, bachelor's or master's

Research Scientist, Life Sciences (experimental)

Molecular biology and biochemistry experiments; PhD required

Research Scientist, Life Sciences (computational)

Analysis of biological data alongside experimental scientists

AI and biology meet in protein design

2024 Nobel Prize in Chemistry

David Baker, for computational protein design

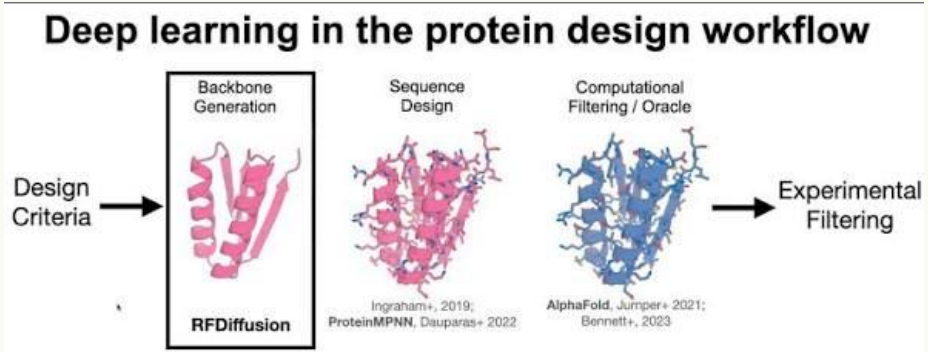
Demis Hassabis and John Jumper, for AlphaFold structure prediction

Source: Anthropic careers postings, 2026; Nobel Prize in Chemistry 2024

GenAI and Protein Engineering

Nobel Prize in Chemistry 2018

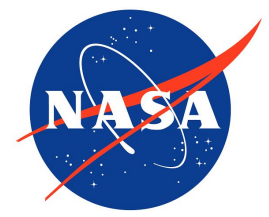
- Frances H. Arnold: Developed the directed evolution of enzymes to create catalysts with improved properties.
- Further improved upon by introduction of de novo protein design by Prof. David Baker
- Proteins are enzymes and catalysts which are the heart of many industrial processes (especially in pharma)



Both experimental and computational expertise are needed for protein scaffold design

Source: Nobel Prize in Chemistry

NASA needs synthetic biologists too!



Make it there, not take it there

NASA Space Synthetic Biology project

BioNutrients

Yeast engineered to make nutrients on demand; a five-year experiment on the International Space Station from 2019

CO2-based manufacturing

Turns CO2 and water into feedstock for microbes that make food, medicines and plastics

Where it sits

Space Synthetic Biology project at NASA Ames Research Center, since 2018

Source: NASA Space Synthetic Biology project, Ames Research Center

SynBio research hubs in India

tifr

ncbs
National Centre for Biological Sciences
Tata Institute of Fundamental Research

iitm research park

WRCB
Supported by WADHWANI FOUNDATION
Wadhwani Research Centre for Bioengineering (WRCB)
IIT Bombay

Entry routes: PhD and integrated PhD programs, research assistant and project roles, postdocs, faculty careers

In deep tech like SynBio, academia and industry work side by side

HaystackAnalytics

- Founded in 2018 and incubated at SINE, IIT Bombay
- Sequencing-based tests for infectious diseases such as tuberculosis and sepsis
- Co-founded with Prof. Kiran Kondabagil

ClarityBio

- Founded in 2020 by Prof. Pramod Wangikar
- Emzing: A Deep Learning-Enabled Software Suite for End-to-End Analysis of High-Resolution LC-MS Metabolomics Data

Source: HaystackAnalytics; SINE, IIT Bombay

iGEM SynBio Job Board

Academia

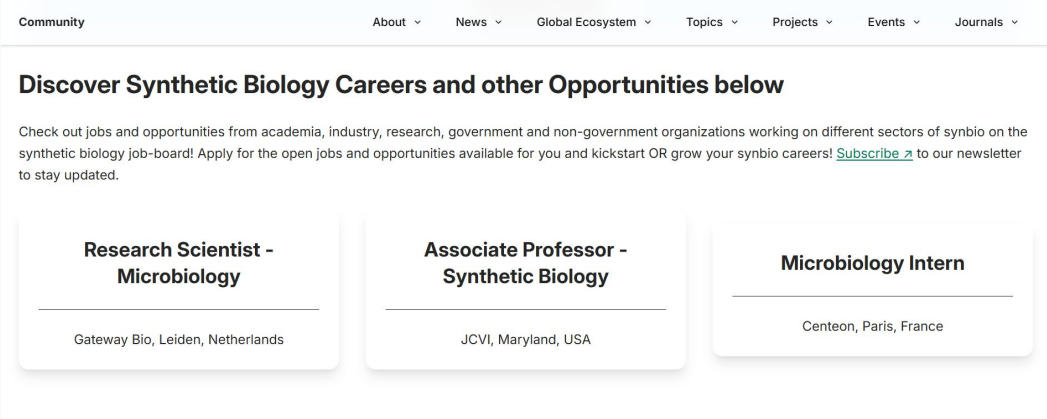
Labs and research groups

Industry

Sponsors and industry partners

iGEM startups

Young companies in the iGEM community



https://community.igem.org/job-board

Opportunities are growing, but SynBio still has open challenges

Opportunities

- Active in academia, industry, AI labs and space agencies
- Roles on both the dry-lab and wet-lab sides
- Research hubs in India and startups at IITB
- Entry routes through iGEM, internships and the job board

Challenges

- Scaling lab results to industrial production
- Reproducibility and standard parts across labs
- Biosafety and biosecurity rules
- Training that spans biology, engineering and data

Questions

Kadambari Bhide and Husayn Bootwala



Appendix

Backup slides



Five career paths, each with its own entry route

Path	Typical roles	Usual entry route
Academic research	PhD student, postdoc, professor	MS or PhD, then a postdoc
Biotech and pharma R&D	Research scientist, process development engineer	MS or PhD for most research roles
Startups	Founding scientist, product developer	Projects, iGEM, incubator experience
Computational biology and automation	Bioinformatician, ML scientist, biofoundry engineer	Coding plus biology projects
Policy, IP and consulting	Biosecurity analyst, patent analyst, biotech consultant	Domain knowledge, then law, policy or MBA training

Every path draws on four skills: go deep in one, stay fluent in the rest



Wet-lab skills

Cloning, cell culture, assays and sterile technique

Matters most in:
academic research, biotech R&D



Coding and data

Python, data analysis and sequence tools

Matters most in:
computational biology, automation



Quantitative modelling

Statistics, reaction kinetics and design of experiments

Matters most in:
bioprocess work, computational biology



Communication

Reading papers, writing and teamwork across fields

Matters most in:
startups, policy, consulting

Build foundations early, test options in year 3, commit in your final year

Years 1–2: Build

- Take core biology, programming and statistics courses
- Volunteer in a lab or join a student team
- Read three review papers on SynBio

Year 3: Test

- Do a summer internship in a lab or company
- Run one project, such as an iGEM-style build
- Try one computational and one wet-lab task

Final year: Commit

- Choose a SynBio topic for your BTP
- Decide between MS or PhD, industry and a startup
- Apply to programs or jobs in the first semester



Next steps

1

Pick one skill and start it

Python for data work, or a molecular biology course or lab module

2

Talk to two people in the field

A senior or alum working in SynBio, and a faculty member whose work interests you

3

Join one team or project

A student team, a volunteer spot in a lab, or a summer internship application

